Homework 5: Ontology-based Semantic Grounding Group 06

Semantic Grounding Ontology

Group 06

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1 Introduction

This report describes the Group 06 ontology-based semantic grounding artifact for Homework 5. The goal of this homework is to construct a small but semantically explicit ontology for a robot agent with grasping capability. The ontology connects perceived or simulated task objects with object types, task roles, manipulation affordances, OWL-based semantic definitions, inferred graspability, and SPARQL query results.

Our group focuses on the cutlery arrangement task from the AI Capstone course project. In this task, a robot arranges a knife and a fork with respect to a plate. Although cutlery arrangement is the main task context, the ontology also includes the required baseline objects from the other predefined course tasks: cup stacking and toy block collection.

The central semantic question is:

> Which objects in the environment are graspable, and why?

The answer is not manually written as a flat list of graspable objects. Instead, graspability is derived from object classes, affordance definitions, and OWL reasoning.

2 Repository Contents

The repository follows the required Homework 5 structure. The main files and folders are:

- `README.md`: explains the repository structure, ontology design, namespace usage, imported resources, reasoning workflow, query execution, and expected results.
- `report.pdf`: this report, explaining the ontology design rationale, reused and newly introduced terms, key axioms, reasoning pattern, query results, design choices, and limitations.
- `ontology/group-ontology.ttl`: the group-authored ontology.
- `ontology/inferred-results.ttl`: the inferred graph generated after reasoning.
- `ontology/imports/course-affordance.ttl`: the provided course ontology used as the shared vocabulary.
- `queries/graspable_objects.rq`: the required SPARQL query for retrieving inferred graspable objects.
- `results/graspable_objects_output.txt`: saved output of the graspable-object query.
- `results/screenshots/`: screenshots of reasoning and query verification.
- `src/`: optional scripts or configuration files used for reproducibility.

3 Namespace Policy

The shared course ontology uses the namespace:

```
@prefix cap: <https://hcis.io/ontology/aicapstone/2026/> .
```

The Group 06 ontology uses the namespace:

```
@prefix g06: <https://hcis.io/ontology/aicapstone/2026/group06/> .
```

The `cap:` namespace is reserved for shared course-level vocabulary, such as physical objects, manipulation tasks, affordance classes, task roles, and predefined object classes. The `g06:` namespace is used for group-specific task instances and object individuals.

This separation prevents group-specific instances from being incorrectly placed inside the shared course namespace.

4 Reused and Newly Introduced Terms

The group ontology reuses the shared course ontology vocabulary for major robotics concepts. Important reused terms include physical object, manipulation task, affordance, grasping affordance, support affordance, containment affordance, stackability affordance, target object, reference object, collectable object, container target, cup, knife, fork, plate, toy block, basket, and graspable object.

The group ontology introduces scene-specific individuals for the AI Capstone tabletop environment. These include the knife, fork, plate, blue cup, pink cup, toy blocks, and basket used in the task setting.

Object Instance	Object Class	Task Role	Inferred Graspable?
`blueCup01`	Cup	TargetObject	Yes
`pinkCup01`	Cup	ReferenceObject	Yes
`knife01`	Knife	TargetObject	Yes
`fork01`	Fork	TargetObject	Yes
`plate01`	Plate	ReferenceObject	No
`block01`	ToyBlock	CollectableObject	Yes
`block02`	ToyBlock	CollectableObject	Yes
`block03`	ToyBlock	CollectableObject	Yes
`basket01`	Basket	ContainerTarget	No

Table 1: Modeled task objects, semantic roles, and expected graspability results.

5 Task and Object Modeling

The main selected task is cutlery arrangement. In this task, the knife and fork are modeled as direct manipulation targets, while the plate is modeled as a reference object used for placement and arrangement.

The required baseline objects for cup stacking and toy block collection are also included. The blue cup is modeled as a target object, while the pink cup is modeled as a reference object. The toy blocks are modeled as collectable objects, and the basket is modeled as a container target.

This modeling separates task relevance from graspability. A task-relevant object is not automatically considered graspable. For example, the plate is important for cutlery arrangement because it provides a spatial reference, but it is not modeled as a direct grasp target. Similarly, the basket is important for toy block collection because it receives the blocks, but it is not modeled as the object being collected.

6 Key Axiom and Reasoning Pattern

The central reasoning target is `cap:GraspableObject`. The intended Description Logic definition is:

$$\begin{aligned}\texttt{cap:GraspableObject} \equiv\ & \texttt{cap:PhysicalObject} \\ & \sqcap\ \exists\ \texttt{cap:hasAffordance.cap:GraspingAffordance}\end{aligned}$$

This means that an object can be classified as graspable if it is a physical object and has at least one grasping affordance.

In Turtle, the pattern is represented using an OWL equivalent-class definition:

```
cap:GraspableObject
    a owl:Class ;
    rdfs:label "graspable object"@en ;
    rdfs:comment
        "A physical object that has at least one grasping affordance and can therefore
            be inferred as graspable."@en ;
    owl:equivalentClass [
        a owl:Class ;
        owl:intersectionOf (
            cap:PhysicalObject
            [
                a owl:Restriction ;
                owl:onProperty cap:hasAffordance ;
                owl:someValuesFrom cap:GraspingAffordance
            ]
        )
    ] .
```

This axiom makes graspability inferable. The final graspable-object classification is not manually asserted on the object instances.

7 Asserted Facts versus Inferred Results

The ontology asserts object types, task roles, labels, pose frames, and affordance-related information. For example, the fork instance is asserted as a fork and assigned the target-object role in the cutlery arrangement task.

The ontology does not manually assert the final conclusion:

```
g06:fork01 a cap:GraspableObject .
```

Instead, this result is inferred through the reasoning pattern. The reasoning chain is:

1. The fork instance is an instance of the Fork class.
2. Fork is a subclass of PhysicalObject.
3. Fork is associated with a grasping-affordance restriction.
4. GraspableObject is equivalent to a physical object with some grasping affordance.
5. Therefore, the fork instance is classified as a graspable object.

The same reasoning pattern applies to the knife, cups, and toy blocks.

8 Reasoning Workflow

The ontology was checked using Protégé with an OWL reasoner. For visual verification, a combined test ontology was loaded into Protégé. After starting the reasoner, the classes Cup, Fork, Knife, and ToyBlock were classified under GraspableObject.

The reasoning result confirms that the ontology supports class-based semantic inference. For example, Cup is recognized as a subclass of GraspableObject, and the blue cup and pink cup are instances of Cup. Therefore, the cup instances are graspable through the inferred class hierarchy.

The inferred graph was exported and saved as:

```
ontology/inferred-results.ttl
```

This inferred graph is used as the data source for SPARQL query execution.

9 SPARQL Query Result

The required SPARQL query is stored in:

```
queries/graspable_objects.rq
```

The query retrieves individuals typed as `cap:GraspableObject`:

```
PREFIX cap: <https://hcis.io/ontology/aicapstone/2026/>
PREFIX g06: <https://hcis.io/ontology/aicapstone/2026/group06/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?obj ?label ?role
WHERE {
  ?obj a cap:GraspableObject .
  OPTIONAL { ?obj cap:hasObjectLabel ?label . }
  OPTIONAL { ?obj cap:hasTaskRole ?role . }
}
ORDER BY ?obj
```

The expected query output includes the two cups, the knife, the fork, and the three toy blocks:

- `blueCup01`
- `pinkCup01`
- `knife01`
- `fork01`
- `block01`
- `block02`
- `block03`

The expected output does not include:

- `plate01`
- `basket01`

This is because the plate is modeled as a reference/support object, and the basket is modeled as a container target. They are task-relevant, but they are not direct grasp targets under our modeling choice.

The saved query output is included in:

```
results/graspable_objects_output.txt
```

10 Design Choices

The main design choice is to distinguish object type, task role, affordance, object instance, and inferred class membership.

For example, the plate is modeled as a Plate and assigned the ReferenceObject role. It is task-relevant because it provides a reference for arranging the knife and fork. However, it is not classified as a graspable object because the ontology models it with support affordance rather than grasping affordance.

Similarly, the basket is modeled as a Basket and assigned the ContainerTarget role. It is important for the toy block collection task, but it is modeled as a container target rather than a collected object. Therefore, it is not returned by the graspable-object query.

This avoids the common modeling error of treating every task-relevant object as graspable.

11 Relationship to the Robot Learning Pipeline

The semantic layer does not replace the robot learning model. Instead, it complements the learning pipeline by making task-object meanings explicit, inspectable, and queryable.

A learned policy may operate over observations, object poses, and actions. The ontology provides an additional symbolic layer that can explain why some objects are considered direct manipulation targets and why other objects are treated as references or containers.

For example, if the policy moves the fork, the ontology can explain that the fork is a direct manipulation target and is semantically graspable. If the policy uses the plate as an arrangement reference, the ontology can explain that the plate is task-relevant but not a direct grasp target in this model.

12 Limitations

This ontology is intentionally compact and educational. It does not model all physical or geometric conditions required for real grasp success.

The current ontology does not represent:

- exact object geometry,
- gripper width constraints,
- object mass,
- friction,
- collision checking,
- grasp pose stability,
- task success probability,
- learned policy confidence,

- temporal action sequences.

Therefore, GraspableObject should be interpreted as semantic graspability in the course ontology, not as a guarantee that a physical robot can successfully grasp the object in every possible state.

Future extensions could add gripper-specific constraints, object dimensions, SHACL validation rules, or links between semantic graspability and policy success or failure cases.

13 Conclusion

This homework demonstrates an ontology-based semantic grounding loop for the AI Capstone tabletop manipulation environment. The group ontology connects observed task objects to shared course object classes, task roles, affordance concepts, and OWL reasoning patterns.

The main result is that the knife, fork, cups, and toy blocks are inferred as graspable objects, while the plate and basket remain task-relevant but non-graspable under the current modeling choices.

This makes the robot task environment more inspectable. Instead of only having visual labels or simulation names, the objects are grounded as typed, queryable, and inferable semantic entities.